

Enrico GIRALDI

Ph.D. Researcher | Computational Neuroscience

 [github.com/github.com/EnricoGiraldi](https://github.com/EnricoGiraldi)

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 Roma, Italia



I am currently a PhD student at ISTC-CNR, the Institute of Cognitive Sciences and Technologies of the Italian National Research Council. My research focuses on the use of EEG (electroencephalography) within the predictive coding framework to understand cognitive processes. My work explores how the human brain anticipates and interprets sensory information by analyzing the underlying neural mechanisms. Additionally, I am interested in the development of probabilistic generative models to study the representation of the external world in the human brain.

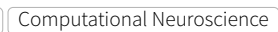
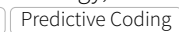
PROFESSIONAL EXPERIENCE

2023 - On going
Roma, Italia

Ph.D. Theoretical And Applied Neuroscience

Università degli studi di Camerino - Consiglio Nazionale delle Ricerche (ISTC-CNR)

My research activity is focused on investigating cognitive aspects at both behavioral and neurophysiological levels, using EEG (electroencephalography) within the theoretical framework of predictive coding. My research aims to understand how the human brain anticipates and interprets sensory information, exploring the underlying neural mechanisms, behavioral implications, and influences on interpersonal and social behavior. Through advanced methodologies and EEG data analysis, I am investigating predictive processes and their relationships with perception, attention, and memory. This research not only contributes to understanding the neurobiological foundations of human cognition but also holds potential applications in fields such as artificial intelligence, neurotechnology, and social neuroscience.

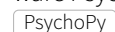
     

03/2025 - On going
Roma, Italia

Tutor

Università di Tor Vergata

Tutor with students in the faculty of psychology. Set up face-to-face and hands-on lectures on using software PsychoPy and Matlab to construct a emotion face recognition experiment in cognitive psychology.

07/2022 - 31/2022
Roma, Italia

Psychiatric Rehabilitation Technician

Villa Maria Pia, casa di cura

I have actively participated in and contributed to rehabilitation groups for patients with severe psychiatric disorders, working within interdisciplinary research teams to enhance patient care and treatment approaches. This collaborative experience has enriched my ability to integrate diverse perspectives and methodologies, furthering both my research and applied skills in clinical settings.



03/2024 - 07/2024
Roma, Italia

Tutor

Università di Tor Vergata

Tutor with students in the faculty of psychology. Set up face-to-face and hands-on lectures on using Matlab to construct a lexical decision experiment in cognitive psychology.

09/2020 - 05/2022
Roma, Italia

Research Psychologist
Istituto di Ortofonia

Collaborated on a research protocol using electroencephalography (EEG) with children on the autism spectrum to assess how treatment impacts brain activity. This project involved close teamwork with a multidisciplinary group of clinicians, psychologists, and researchers, fostering a supportive and socially-focused environment to optimize data collection and EEG analysis. Our collective effort aimed not only at understanding neurological outcomes but also at addressing broader social and behavioral benefits for the children and their families.

Supervisor : Dr. Giancarlo Zito.

EEG EEGLab Matlab

07/2018 - 31/2020
Roma, Italia

Research Psychologist
Fondazione Santa Lucia

Evaluation of an EEG-based Brain-Computer Interface (BCI) system for communication in clinical settings." This work was conducted within interdisciplinary research teams, where I collaborated closely with specialists from various fields to assess the system's efficacy and usability for patient communication. These collaborative efforts enriched the research process, allowing for comprehensive insights and a well-rounded approach to clinical application. - "DoCMA Project" : Evaluation of a BCI with patients with disorders of consciousness. During the DoCMA project, I had the opportunity to work abroad :

- > 03/2019 - 05/2019, Visiting Researcher : Vithas Nisa Hospital, Seville, Spain - Clinical assessment of patients with disorders of consciousness
- > 09/2019 - 12/2019, Visiting Researcher : g.tec medical engineering, Sierningstraße, Austria - Analysis of EEG data recorded with the mindBEAGLE BCI system

Supervisor : Dr. Donatella Mattia

EEG BCI Matlab SPSS Statistica

03/2017 - 09/2017
Roma, Italia

Professional Educator
La Sponda cooperativa sociale

I have worked on the following projects, assisted children in socialization and integration within group settings.

- > SISMIFF involves young people living in challenging social environments
- > SAISH involves children with physical or mental disabilities

Social work developmental age

 EDUCATION

01/2023 - in corso
Roma, Italia

42 programming school
Luiss Buisness School

The common core of the course focuses on learning programming languages through practical projects in C/C++. So far, I have completed projects using the C language. Here are some of the main projects :

- > libft : this project involved creating a C library from scratch with all the functions to be used in subsequent projects.
- > ftprintf : writing a C program that replicates all the operations of the printf function in the C language.
- > solong : creating a program that runs a 2D video game.
- > minishell : creating a C program that runs a shell with all the functions and commands of bash.

01/2021 - 06/2021
Roma, Italia

Data Science Master
Neural Academy

The master's program was focused on two main parts :

- > Introductory course on Python using Jupyter Notebook
- > Use of Git commands and creation of a repository on GitHub to upload personal projects
- > Learning how to write an analysis pipeline to evaluate different machine learning algorithms on various datasets.

2014 - 2016 Roma, Italia	Master's Degree in Cognitive Neuroscience and Psychological Rehabilitation Università degli Studi di Roma La Sapienza Thesis Title : Cognitive Components for the Control of P300-Based Brain-Computer Interfaces Supervisor : Antonino Raffone Discussion Date : 07/2016
2011 - 2014 Roma, Italia	Bachelor's Degree in Psychology and Social Processes Università degli Studi di Roma La Sapienza Thesis Title : Evaluation of Attitudes Toward Violence in Stadiums Through Free Associations Supervisor : Alessandra Areni Discussion Date : 07/2014

LANGUAGE CERTIFICATIONS

01/2021 **English Language** : TOEFL PBT, Wayne State University (Detroit, Michigan, USA)

SKILLS

Programming and Computer Skills	Python, Matlab, R, C, C++, Shell Script, Networking
Statistics and Data Analysis	SPSS, Statistica, Jamovi, Machine Learning
EEG Signal Analysis and Experimental Psychology	EEGLab, MNE (python), Psychophysic toolbox, Psychopy

PUBLICATIONS

- Angela Riccio, Francesca Schettini, Valentina Galiotta, Enrico Giraldi, Maria Grazia Grasso, Febo Cincotti, Donatella Mattia. 2022. *Usability of a hybrid system combining P300-based brain-computer interface and commercial assistive technologies to enhance communication in people with multiple sclerosis*. Frontiers in Human Neuroscience;
- Angela Riccio, Francesca Schettini, Enrico Giraldi, Federica Cappalonga, Lisa Pelagalli, Febo Cincotti, Donatella Mattia. 2020. *Investigation of Needs and characteristics of End-Users, for a Future Inclusion of BCIs in at-Centers*. GBCIC;